Manuel Berger

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I finished my master's studies, specialising in mathematical physics, at ETH Zürich in September 2025. My thesis was on 'Relations between theta functions on Riemann surfaces of different genera', using the language of the Schottky uniformization, under the supervision of Dr. Johannes Brödel in the String Theory and Quantum Field Theory group of Prof. Niklas Beisert.

Employments

Oct.–Dec. 2025	**Scientific Assistant** *Temporary employment* QFT and Strings group, Institute for Theoretical Physics, ETH Zürich

Education

2023–2025	**ETH Zürich**, CH	
	↦ MSc. Physics, with distinction	GPA 5.78/6
2020–2023	**ETH Zürich**, CH	
	↦ BSc. Physics	GPA 5.21/6
2017–2020	**Technisches Gymnasium Waldshut**, DE	
	↦ Abitur, with distinction	GPA 1.2/1

During my bachelor's studies, I completed dedicated courses on algebra and topology in addition to the mandatory curriculum.

Interests

- Interplay between mathematical fields like complex analysis and abstract algebra in the study of physical concepts.
- Mathematical physics, quantum field theory, string theory, conformal field theory, symmetries, general relativity, cosmology, algebraic topology.
- Implementation, visualisation, and numerical approaches to mathematical frameworks using software such as Mathematica.
- Abstraction and algorithmic problem-solving, designing and implementing efficient numerical routines in Python (NumPy/SciPy) and communicating results with clear visuals (Matplotlib).

Attended conferences, workshops and schools

Sep. 2026	**Elliptics and Beyond '26**. SRS, Les Diablerets, Switzerland. [link]
Apr. 2026	**Hidden Structures in Multi-loop Amplitudes**. Online. [link]
Feb. 2026	**Amplitudes and Algebraic Geometry**. ESI, Vienna, Austria. [link]
Jan. 2026	**Single-Valued Periods in Scattering Amplitudes**. Online. [link]
Nov. 2025	**GeNeZiSS**. Universität Bern, Switzerland. [link]
Sep. 2025	**Elliptics and Beyond '25**. Bethe Center for Theoretical Physics, Bonn, Germany. [link]

Aug. 2025	**MAPSS**, Mathematical Physics Summer School. SRS, Les Diablerets, Switzerland. [link]

Technical skills

Languages	German	Native
	English	Full professional proficiency
IT	Programming	Python, LaTeX, C++
	Software	Mathematica, Git

Awards

2026	ETH Medal for outstanding master's thesis. // CHF 2'000
2020	Abiturpreis and Buchpreis of the German Physical Society.
	Ferry-Porsche-Preis for outstanding grades in mathematics and physics.

Articles and preprints

- M. Berger, J. Broedel, *'Translating auxiliary symmetries between Schottky uniformization and Jacobi parametrization'*, 2026. [J. Phys. A: Math. Theor.] [arXiv:2602.04715]
  Based in part on my master's thesis at ETH Zürich.

Academic references

- **Dr. Johannes Broedel**,
  Institute for Theoretical Physics, ETH Zürich.
  Email: jbroedel@ethz.ch
- **Dr. Stephan Stieberger**,
  Max Planck Institute for Physics, Garching.
  Email: stephan.stieberger@mpp.mpg.de